

$$\begin{array}{l} \text{Row 3} = \text{Row 3} \\ - \text{Row 2} \end{array} \rightarrow \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 11 & -2 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

Verification of whether the final matrix is in echelon form

1. There are no zero rows in the matrix. $\therefore$ The condition that all non-zero rows must be above all the zero rows is vacuously True.

2. Row 1 Leading entry = 1 in Column 1
Row 2 Leading entry = 1 in Column 2
Row 3 Leading entry = -3 in Column 3

$\therefore$ The condition that each row of the matrix is in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

$\therefore$ The condition that for each pivot column, all the elements below the pivot position are 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 11 & -2 \\ 0 & 0 & 0 & -3 \end{bmatrix}$ is in echelon form $\square$

There is a row $[0 \ 0 \ 0 \ -3]$ of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. By Theorem 2, the system of equations corresponding to the augmented matrix $\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 11 & -2 \\ 0 & 0 & 0 & -3 \end{bmatrix}$ is inconsistent, i.e. no solution exists.

$\therefore$ The eqn. $Ax=c$ has no solution $\square$